

BABRUYSK, Belarus

WMO: 269610

Lat: 53.2085N

Lon: 29.1273E

Elev: 513

StdP: 14.43

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	0.0	-11.3	3.1	-6.4	-4.8	4.3	0.7	22.3	35.6	20.0	36.1	3.7	30	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	85.0	67.6	81.8	66.5	78.6	65.1	69.9	81.2	68.3	78.2	66.6	75.8	6.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	98.4	75.0	64.7	93.2	72.7	63.2	88.4	70.7	34.1	81.6	32.8	78.2	31.5	76.0	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	16.4	14.6	DB	-15.1	89.6	5.0	4.1	-18.7	92.5	-21.6	94.9	-24.4	97.2	-28.0	100.1	(4)
(5)			WB	-15.4	72.8	4.9	1.8	-19.0	74.1	-21.9	75.1	-24.7	76.1	-28.3	77.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.5	23.0	24.6	32.9	45.9	55.3	61.7	66.1	63.7	54.0	43.9	34.8	26.5		
	DBStd	17.27	11.17	11.08	8.48	7.67	7.78	6.21	5.47	5.95	6.43	7.48	8.85	10.67		
	HDD50	3720	838	711	530	168	34	0	0	0	34	218	459	728		
	HDD65	7688	1303	1131	994	574	313	139	52	96	331	655	907	1193		
	CDD50	1705	0	0	1	45	200	351	500	425	154	28	1	0		
	CDD65	197	0	0	0	0	13	39	87	56	2	0	0	0		
	CDH74	1929	0	0	0	9	153	368	829	534	36	0	0	0		
	CDH80	486	0	0	0	1	27	67	230	160	1	0	0	0		
(14) Wind	WSAvg	6.6	7.3	7.4	7.6	6.7	6.5	6.3	5.5	5.5	5.5	6.6	6.9	7.3		
(15) Precipitation	PrecAvg	25.1	1.6	1.5	1.5	1.7	2.4	2.9	3.6	2.2	2.0	2.2	1.8	1.7		
	PrecMax	33.0	3.1	3.3	2.9	3.3	4.5	6.3	6.8	5.0	5.0	6.6	3.8	3.0		
	PrecMin	19.3	0.3	0.6	0.2	0.1	0.5	1.0	0.5	0.2	0.2	0.1	0.1	0.5		
	PrecStd	3.4	0.6	0.6	0.7	0.8	1.1	1.3	1.7	1.2	1.2	1.6	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.2	46.2	59.4	74.3	83.6	85.8	90.0	92.7	78.7	69.8	54.9	47.4		
		MCWB	41.7	41.7	47.7	55.2	65.3	68.0	70.3	69.6	62.1	58.7	51.6	45.2		
	2%	DB	39.9	41.7	53.8	69.0	78.3	82.2	85.6	83.7	73.7	63.0	51.6	44.0		
		MCWB	38.2	38.9	44.6	53.2	63.6	66.4	68.0	67.8	60.8	55.7	49.0	42.6		
	5%	DB	37.7	38.9	48.8	64.4	73.9	78.7	82.3	79.5	69.3	58.5	48.3	41.3		
		MCWB	36.1	37.0	41.9	51.2	60.3	64.8	67.3	65.2	58.9	53.1	46.2	40.0		
	10%	DB	35.1	36.6	44.6	60.0	70.1	74.9	78.7	75.7	65.6	55.1	45.7	38.1		
		MCWB	34.0	34.8	40.1	49.3	57.9	63.2	66.1	63.9	57.1	51.7	44.1	36.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.2	43.0	49.2	57.9	67.7	70.1	72.8	71.3	64.1	61.3	52.5	45.7		
		MCDB	43.8	45.9	58.0	71.0	79.5	81.3	84.9	83.4	75.1	67.6	54.3	46.6		
	2%	WB	38.5	39.5	45.5	54.9	64.6	68.1	70.7	69.3	62.2	56.9	49.4	43.0		
		MCDB	39.6	41.4	51.4	64.1	75.6	78.4	81.8	81.0	71.0	60.9	51.3	43.8		
	5%	WB	36.3	37.2	42.8	52.8	62.2	66.5	68.7	67.6	60.1	54.1	46.5	40.1		
		MCDB	37.5	38.7	47.9	61.6	71.7	76.2	78.4	76.8	66.6	57.8	48.2	41.3		
	10%	WB	34.2	35.1	40.1	50.8	59.8	64.6	66.9	65.6	58.3	51.6	44.1	36.9		
		MCDB	35.1	36.4	44.1	58.7	67.8	72.7	76.0	73.1	63.7	54.9	45.5	38.0		
(35) Mean Daily Temperature Range	MDBR		8.4	10.0	13.3	18.2	19.5	18.6	19.2	19.8	18.3	13.4	7.9	7.4		
	5% DB	MCDBR	6.7	9.7	20.0	25.9	25.4	24.5	24.4	25.5	24.3	19.1	11.0	7.7		
		MCWBR	6.3	7.7	12.2	12.5	11.9	11.0	9.7	10.8	13.0	11.9	8.6	6.9		
	5% WB	MCDBR	6.4	8.7	17.5	21.8	23.0	20.7	21.6	21.5	20.5	16.2	9.7	7.7		
		MCWBR	6.3	7.3	11.6	11.3	11.6	10.5	10.1	10.0	11.9	10.8	7.9	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.329	0.361	0.417	0.404	0.390	0.417	0.424	0.384	0.355	0.337	0.299		
	taud		2.157	2.105	2.155	2.171	2.284	2.351	2.286	2.259	2.329	2.386	2.273	2.198		
	Ebn at Noon		208	237	257	257	268	273	263	252	249	231	193	183		
	Edh at Noon		22	31	37	42	39	37	39	38	31	24	20	17		
(44) All-Sky Solar Radiation	RadAvg		225	481	906	1302	1664	1804	1716	1466	989	518	187	134		
	RadStd		44	95	137	154	145	137	156	130	96	96	26	37		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days					
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)		
(47)	Regional (10 neighbors)	+1.08	N/A	N/A	N/A	N/A	+1.35	N/A	-328	+184	N/A	(47)		

Nomenclature: See separate page